

Assignment 10.3 - Series

Evaluate the geometric series.

$$1. \sum_{k=0}^{\infty} \left(\frac{2}{5}\right)^k \rightarrow \left|\frac{2}{5}\right| < 1 \rightarrow \text{converges} \rightarrow a=1, r=\frac{2}{5}$$

$$\sum_{k=0}^{\infty} \left(\frac{2}{5}\right)^k = \frac{1}{1-\frac{2}{5}} = \boxed{\frac{5}{3}}$$

$$2. \sum_{k=0}^{\infty} \left(-\frac{7}{10}\right)^k \rightarrow \left|-\frac{7}{10}\right| < 1 \rightarrow \text{converges} \rightarrow a=1, r=-\frac{7}{10}$$

$$\sum_{k=0}^{\infty} \left(-\frac{7}{10}\right)^k = \frac{1}{1-\frac{-7}{10}} = \boxed{\frac{10}{17}}$$

$$3. 1 + \frac{2}{7} + \frac{2^2}{7^2} + \frac{2^3}{7^3} + \dots \rightarrow \left|\frac{2}{7}\right| < 1 \rightarrow \text{converges} \rightarrow a=1, r=\frac{2}{7}$$

$$\sum_{k=0}^{\infty} \left(\frac{2}{7}\right)^k = \frac{1}{1-\frac{2}{7}} = \frac{1}{\frac{5}{7}} = \frac{7}{5}$$

$$4. \sum_{n=0}^{\infty} e^{-6n}$$

$$\sum_{n=0}^{\infty} \frac{1}{e^{6n}} = \sum_{n=0}^{\infty} \left(\frac{1}{e^6}\right)^n \rightarrow \left|\frac{1}{e^6}\right| < 1 \rightarrow \text{converges} \rightarrow \sum_{n=0}^{\infty} e^{-6n} = \frac{1}{1-\frac{1}{e^6}} = \frac{e^6}{e^6-1}$$

For the following telescoping series, find a formula for the n^{th} term of the sequence of partial sums $\{S_n\}$.Then evaluate $\lim_{n \rightarrow \infty} S_n$ to obtain the value of the series or state that it diverges.

$$5. \sum_{k=1}^{\infty} \left(\frac{10}{k} - \frac{10}{k+1}\right)$$

$$\begin{aligned} k=1 & 10 - \frac{10}{2} \\ k=2 & \frac{10}{2} - \frac{10}{3} \\ k=3 & \frac{10}{3} - \frac{10}{4} \\ k=4 & \frac{10}{4} - \frac{10}{5} \\ k=n & \frac{10}{n} - \frac{10}{n+1} \end{aligned}$$

$$S_n = 10 - 5 + 5 - \frac{10}{3} + \frac{10}{3} - \frac{10}{4} + \frac{10}{4} - \frac{10}{5} + \dots$$

$$S_n = 10 - \frac{10}{n+1}$$

$$\sum_{k=1}^{\infty} \left(\frac{10}{k} - \frac{10}{k+1}\right) = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(10 - \frac{10}{n+1}\right) = 10 - 0 = \boxed{10}$$

$$6. \sum_{k=1}^{\infty} \frac{15}{(3k-1)(3k+2)}$$

$$A(3k+2) + B(3k-1) = 15$$

$$\text{At } k = -\frac{2}{3}, B = -5$$

$$k = \frac{1}{3}, A = 5$$

$$\begin{aligned} k=1 & \frac{5}{2} - \frac{5}{5} \\ k=2 & \frac{5}{5} - \frac{5}{8} \\ k=3 & \frac{5}{8} - \frac{5}{11} \end{aligned}$$

$$S_n = \frac{5}{2} - \frac{5}{5} - \frac{5}{8} + \frac{5}{8} - \frac{5}{11} + \dots$$

$$S_n = \frac{5}{2} - \frac{5}{3n+2}$$

$$\boxed{\frac{5}{2}}$$

$$7. \sum_{k=1}^{\infty} \left(\frac{2}{\sqrt{k+1}} - \frac{2}{\sqrt{k+3}}\right)$$

$$\begin{aligned} k=1 & \rightarrow \frac{2}{\sqrt{2}} - \frac{2}{\sqrt{4}} \\ k=2 & \rightarrow \frac{2}{\sqrt{3}} - \frac{2}{\sqrt{5}} \\ k=3 & \rightarrow \frac{2}{\sqrt{4}} - \frac{2}{\sqrt{6}} \end{aligned}$$

$$S_n = \left(\frac{2}{\sqrt{2}} + \frac{2}{\sqrt{3}}\right) - \left(\frac{2}{\sqrt{n+2}} + \frac{2}{\sqrt{n+3}}\right) \rightarrow \lim_{n \rightarrow \infty} S_n = \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{3}} - 0 = \boxed{\frac{3\sqrt{2} + 2\sqrt{3}}{3}}$$

$$8. \sum_{k=1}^{\infty} \left(\sin^{-1}\left(\frac{1}{k}\right) - \sin^{-1}\left(\frac{1}{k+1}\right)\right)$$

$$k=1 \quad \sin^{-1}(1) - \sin^{-1}\left(\frac{1}{2}\right)$$

$$k=2 \quad \sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(\frac{1}{3}\right)$$

$$k=3 \quad \sin^{-1}\left(\frac{1}{3}\right) - \sin^{-1}\left(\frac{1}{4}\right)$$

$$k=n \quad \sin^{-1}\left(\frac{1}{n}\right) - \sin^{-1}\left(\frac{1}{n+1}\right)$$

$$S_n = \sin^{-1}(1) - \sin^{-1}\left(\frac{1}{n+1}\right)$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(\sin^{-1}(1) - \sin^{-1}\left(\frac{1}{n+1}\right)\right) = \sin^{-1}(1) - \sin^{-1}(0)$$

$$= \frac{\pi}{2} - 0 = \frac{\pi}{2}$$